Weekly Student Progress Performa

Team Id	CS26084	**Week no / Date**	Week 6 24 August 2026
Project Title	SentinelCore: A Network Detection and Incident Response Platform		
Members Present	Om Rashiya (24cs084), Yash Patel (24cs082)		
Mentor	Akshita Kadam		

WEEK 6 • Asset Discovery and Suricata Sensor

Section	Details to be filled by the team
Planned Tasks (from last week)	Build asset discovery and begin integrating Suricata as the network sensor.
Tasks Completed This Week	Implemented allow-listed helper operations, Nmap scan execution/parsing, asset/port/scan models, scan history, validation, asset detail pages, Suricata configuration, rule sources, rule overrides, health metrics, and safe ruleset reload behavior.
Commits Made	Total: 0 standalone commits this review period. The feature branch was kept as a working milestone and later consolidated in d79a37c (“update the project to m5”).
Issues Created / Closed	Created: Untrusted scan targets and rule text could cross the privilege boundary. Closed: Added strict network/argument validation, allow-listed helper operations, and verify-before-activate rule staging.
Milestone Status	~45% complete — on track. Discovery and sensor administration were operational in development.
Blockers / Challenges Faced	Nmap and Suricata require network access that differs across Linux and macOS Docker hosts. The helper owns capabilities while the API remains isolated.
Plan for Next Week	Normalize EVE JSON, persist events reliably, track offsets, create partitions, and expose pipeline health.

Research and Technical Study

Research objective. Build asset discovery and begin integrating Suricata as the network sensor.

Study completed. Studied Nmap XML parsing, input validation for command execution, Suricata EVE JSON, rule update workflows, AF_PACKET capture, and atomic configuration activation. A staged ruleset is verified before replacing the active set to avoid sensor downtime.

Discovery and sensor capability mapping

Area	Implemented function	Safety or evidence
Asset discovery	Validated Nmap scans	Structured assets, ports, and scan history
Privilege boundary	Allow-listed helper operations	API remains unprivileged
Sensor lifecycle	Status / start / stop	Visible process and health state
Rules	Sources, overrides, staged activation	Verify before reload with rollback
Operations	Throughput and drop counters	Capture health visible to the operator

Decision carried into the project

Implemented allow-listed helper operations, Nmap scan execution/parsing, asset/port/scan models, scan history, validation, asset detail pages, Suricata configuration, rule sources, rule overrides, health metrics, and safe ruleset reload behavior.

Implementation Record and Website Evidence

Discovery execution

Created typed helper operations for allowed scan modes. Targets and arguments are validated before execution, and Nmap XML is parsed into structured scan, asset, and port records.

Inventory behaviour

Assets are reconciled by address, their first/last seen information is retained, discovered ports are associated with the asset, and missing devices can be marked inactive rather than deleted.

Sensor management

Added status, start/stop, rule-source, override, configuration-test, and reload flows. Rule content is staged and tested before activation so an invalid update does not silently replace the working set.

Operational view

The Sensor page exposes status, version, process information, uptime, rules loaded, last reload, throughput/drop indicators, rule sources, and overrides in one workspace.

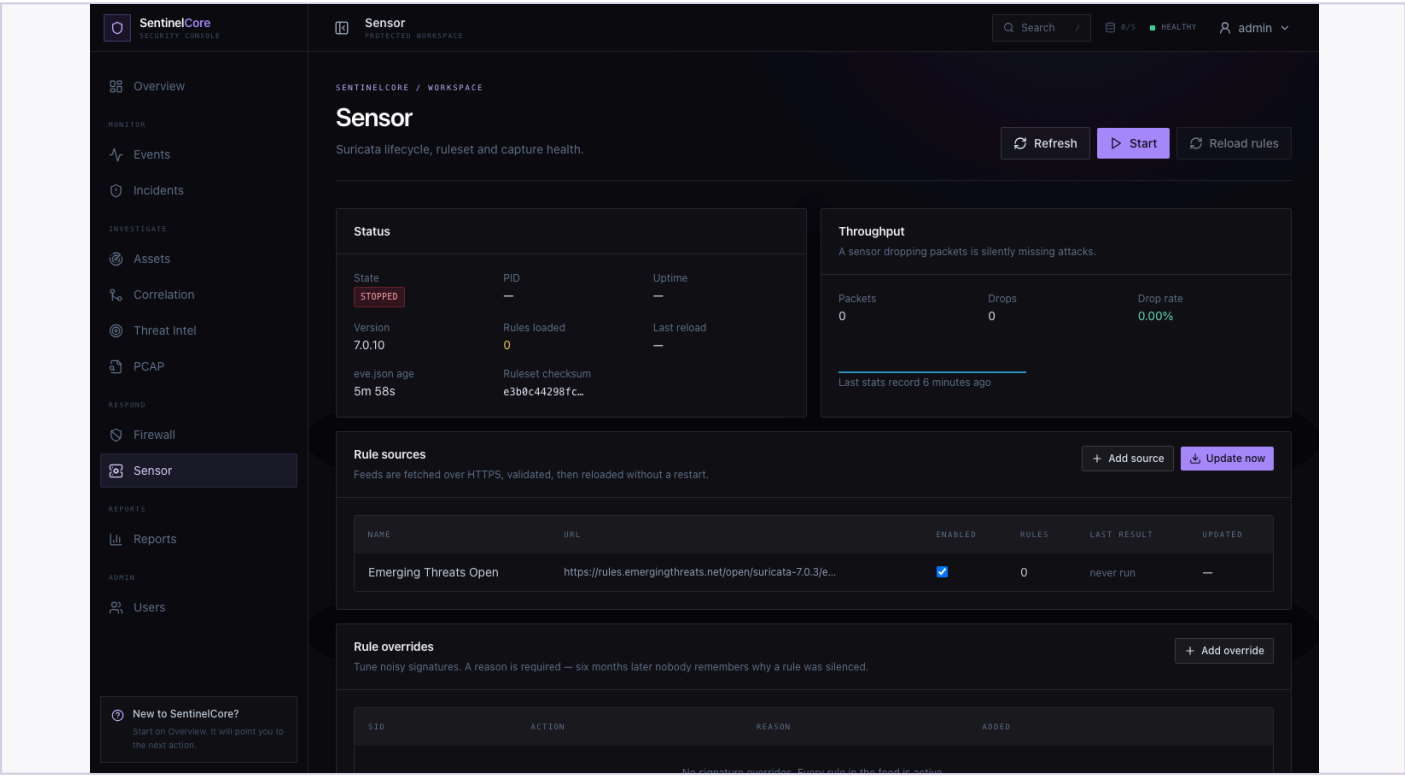


Figure W6.1 — Project-owned SentinelCore interface mapped specifically to Week 6.

Evidence interpretation. The Sensor workspace displays stopped/running state, process and version fields, packet/drop counters, ruleset details, source controls, and rule overrides. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

Repository and Architecture Evidence

Sensor workspace implementation

frontend/src/pages/sensor/SensorPage.jsx

```
  );
}

export default function SensorPage() {
  const toast = useToast();

  const [status, setStatus] = useState(null);
  const [stats, setStats] = useState(null);
  const [events, setEvents] = useState([]);
  const [loading, setLoading] = useState(true);
  const [busy, setBusy] = useState(null);
  const [confirm, setConfirm] = useState(null);

  const refresh = useCallback(async () => {
    try {
      const [s, st, ev] = await Promise.all([
        api.get("/sensor/status"),
        api.get("/sensor/stats").catch(() => null),
        api.get("/sensor/events?limit=15").catch(() => []),
      ]);
      setStatus(s);
      setStats(st);
      setEvents(ev);
    } catch (err) {
```

Repository evidence captured directly from the SentinelCore workspace

Figure W6.2 — Sensor workspace implementation. This is a separate internal artifact, not a copy of the website image.

Repository files used for traceability

- reports/modules/M03-asset-discovery_1.md
- reports/modules/M04-suricata-sensor_1.md
- frontend/src/pages/sensor/SensorPage.jsx
- backend/helper/helper/ops/suricata.py

Verification

Feature names and safety behaviour were checked against the M3/M4 specifications, helper operations, frontend components, and related tests.

Traceability conclusion. Sensor workspace implementation directly supports the Asset Discovery and Suricata Sensor milestone. Feature names and safety behaviour were checked against the M3/M4 specifications, helper operations, frontend components, and related tests. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

Weekly Closure and Faculty Assessment

Completed outcome

Implemented allow-listed helper operations, Nmap scan execution/parsing, asset/port/scan models, scan history, validation, asset detail pages, Suricata configuration, rule sources, rule overrides, health metrics, and safe ruleset reload behavior.

Commit record

Total: 0 standalone commits this review period. The feature branch was kept as a working milestone and later consolidated in d79a37c (“update the project to m5”).

Issue and resolution

Created: Untrusted scan targets and rule text could cross the privilege boundary. Closed: Added strict network/argument validation, allow-listed helper operations, and verify-before-activate rule staging.

Milestone position

~45% complete — on track. Discovery and sensor administration were operational in development.

Challenge handled

Nmap and Suricata require network access that differs across Linux and macOS Docker hosts. The helper owns capabilities while the API remains isolated.

Next-week hand-off

Normalize EVE JSON, persist events reliably, track offsets, create partitions, and expose pipeline health.

Repository reference

https://github.com/rashiyaom/SentinelCore

Assessed By (Faculty)	Marks Awarded (5 marks)	Faculty Remarks